

Chapter 2

HTML Part 3

Links and Navigation

INT102 Web Technologies (1/2023)

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Topics

- What will you learn after finishing this class:
 - internal and external links
 - anchor element
 - absolute and relative URLs
 - Relative URL
 - linking within a directory
 - linking to a lower directory
 - linking to a higher directory
 - target attribute

Hypertext Link (Link)

- [Hypertext Link](#) (or [Hyperlink](#) or [Link](#))
 - is an element in an HTML document that links to either another portion of document or to another document.
 - can be a word, phrase or image.
- [Hyperlink](#) makes the Web different from other mediums in which a web page can contain links.

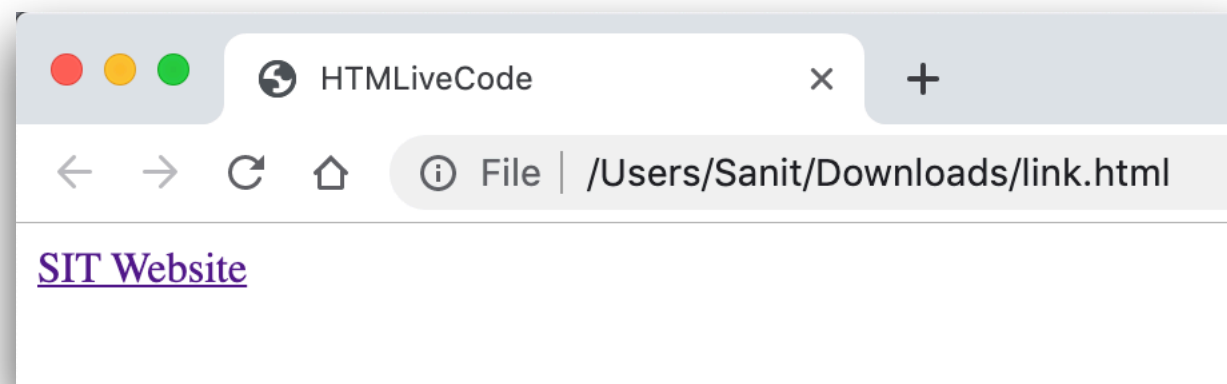
Types of link

- Internal link
 - links to another page or within the page in the same web site of the document.
- External link
 - links to another page in a different site of the document.

Anchor element: <a>

HyperText Link

SIT Website



Absolute URL vs. Relative URL

- Two ways to specify the URL:

- **Absolute URLs**

- provides the full URL for the document, including the protocol (http://), the domain name, and the path name.
- is used to link to another document in the different site.
- eg. "http://www.sit.kmutt.ac.th"

"https://www.tutorialspoint.com/html/html_overview.htm"

- **Relative URLs**

- provides the pathname to a file relative to the current document.
- is used to link to another document in the same site.
- eg. "/html"

"/html/index.html"

"html/resource/"

"../download/lab3.html"

Absolute URL

href="http://domain_name/path_name"

- Linking to Pages on the web

"http://www.sit.kmutt.ac.th"

"https://www.tutorialspoint.com/html/html_overview.htm"

Relative URL

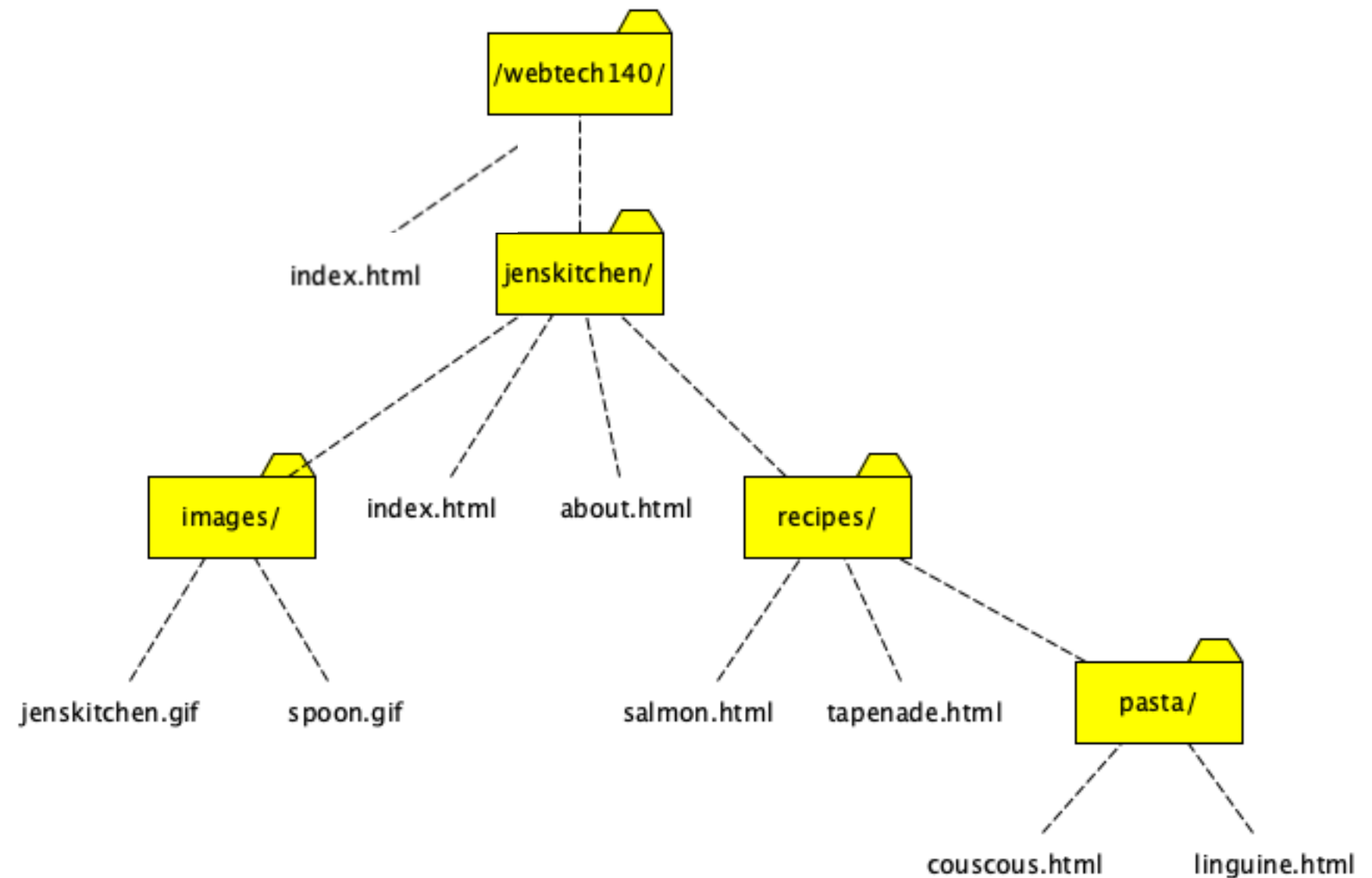
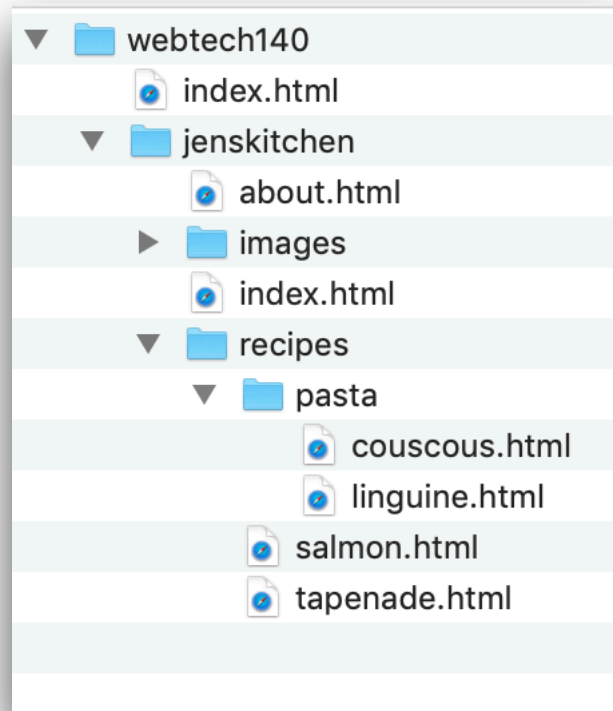
`href="path_name"`

- Use a [Relative URL](#) to link to a page on the [same site](#) (or [server](#))
 - from home page to section pages
 - from section pages to context pages and so on.
- Without "http://", the browser looks on the current server for the linked document.
- [A path_name](#)
 - is used to point to a [particular file](#) or [directory](#).
 - tells the browser where to find the file.
 - follows the Unix convention of separating directory and filenames with [forward slashes \(/ \)](#).

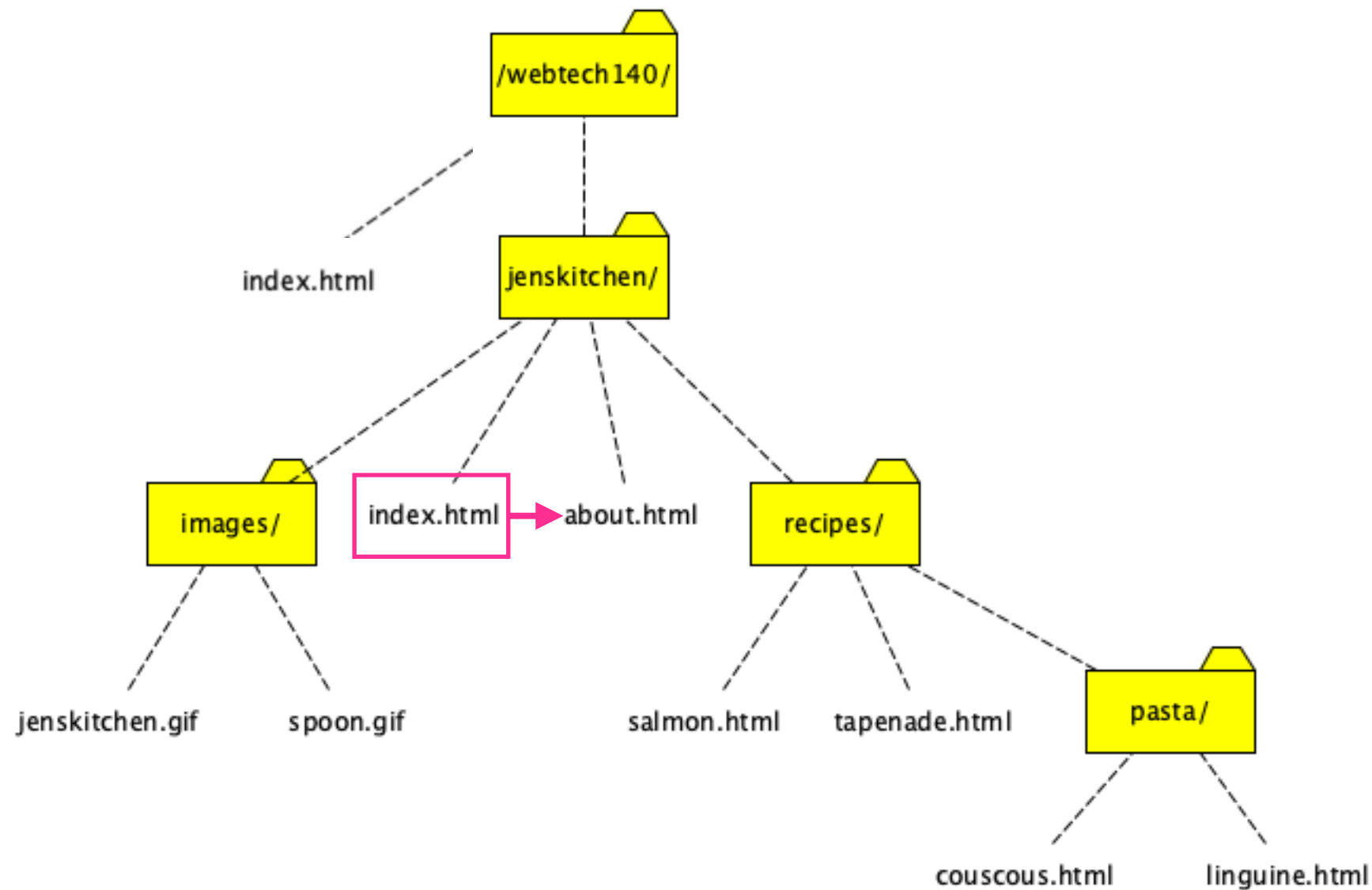
Unix Path Convention

Directory	Description
/	root
.	current directory
..	move up one level
/path	path under the root
path or ./path	path within the current directory
../path	move up one level and look for the path
../../path	move up two levels and look for the path

Site Structure

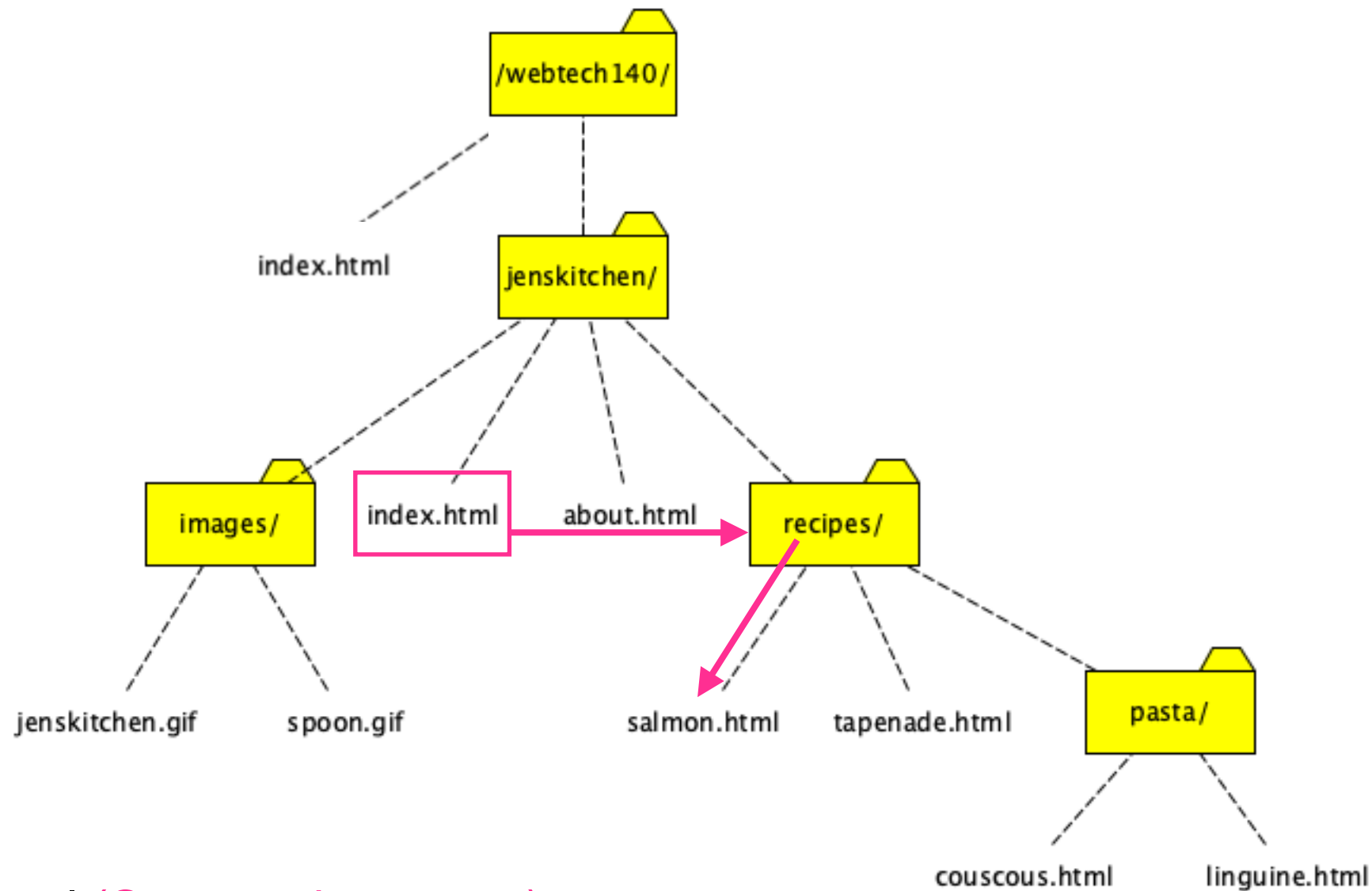


Linking within a directory



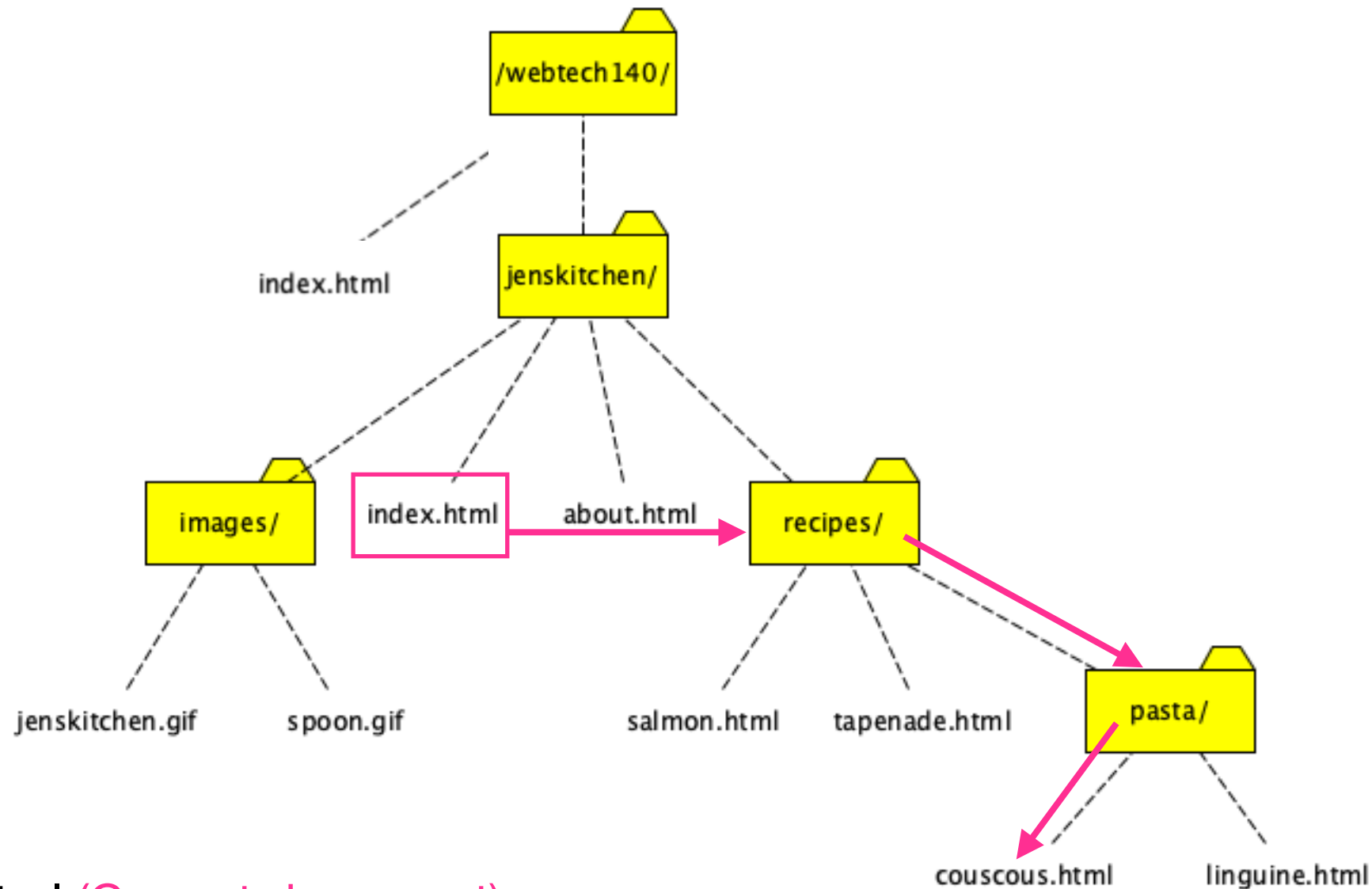
- index.html (Current document)
- `<a href="about.html">About this site</a>`

Linking to a lower directory (one level)



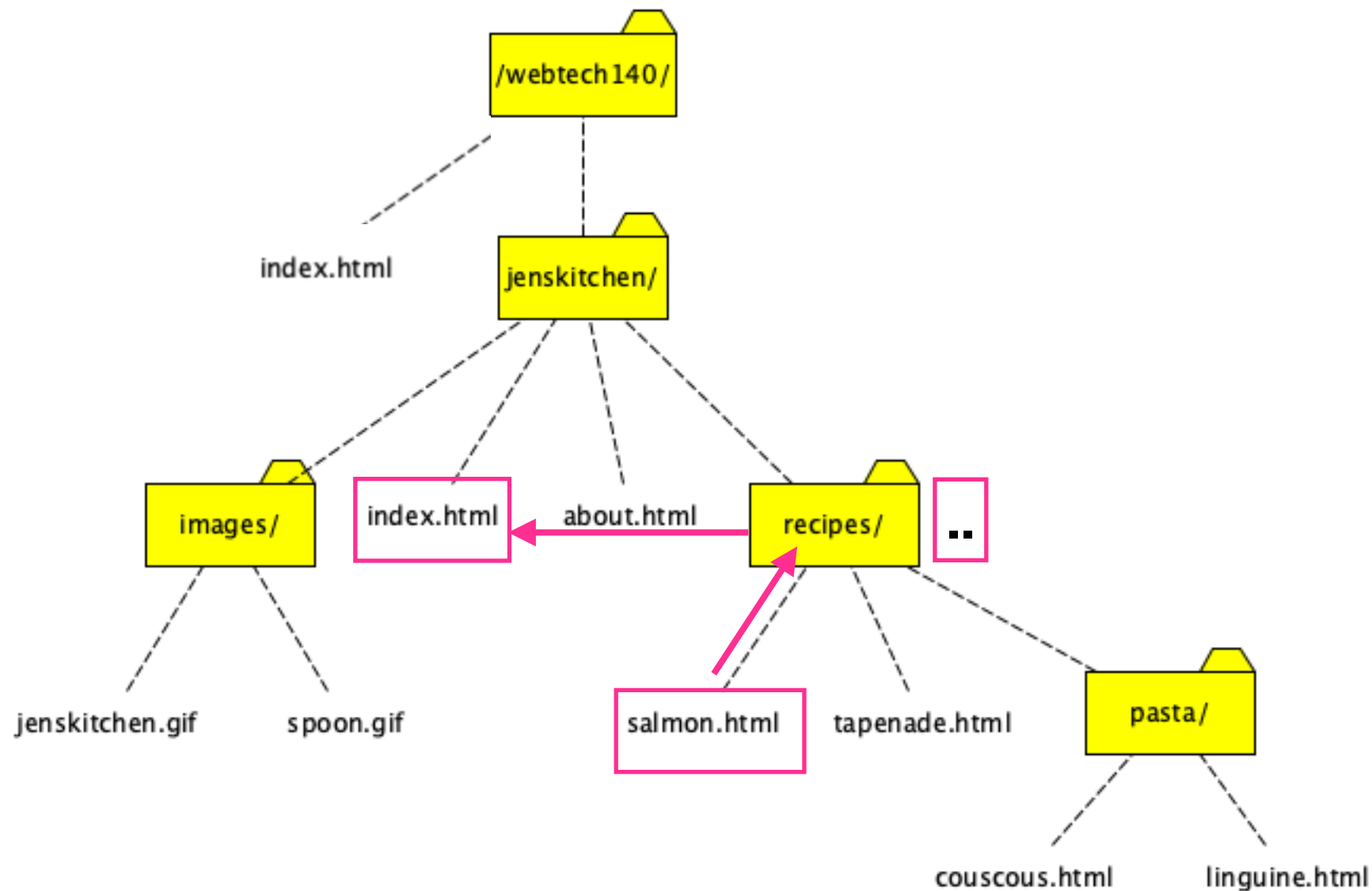
- index.html (Current document)
 - `<a href="recipes/salmon.html">Garlic Salmon</a>` OR
 - `<a href="./recipes/salmon.html">Garlic Salmon</a>`

Linking to a lower directory (two levels)



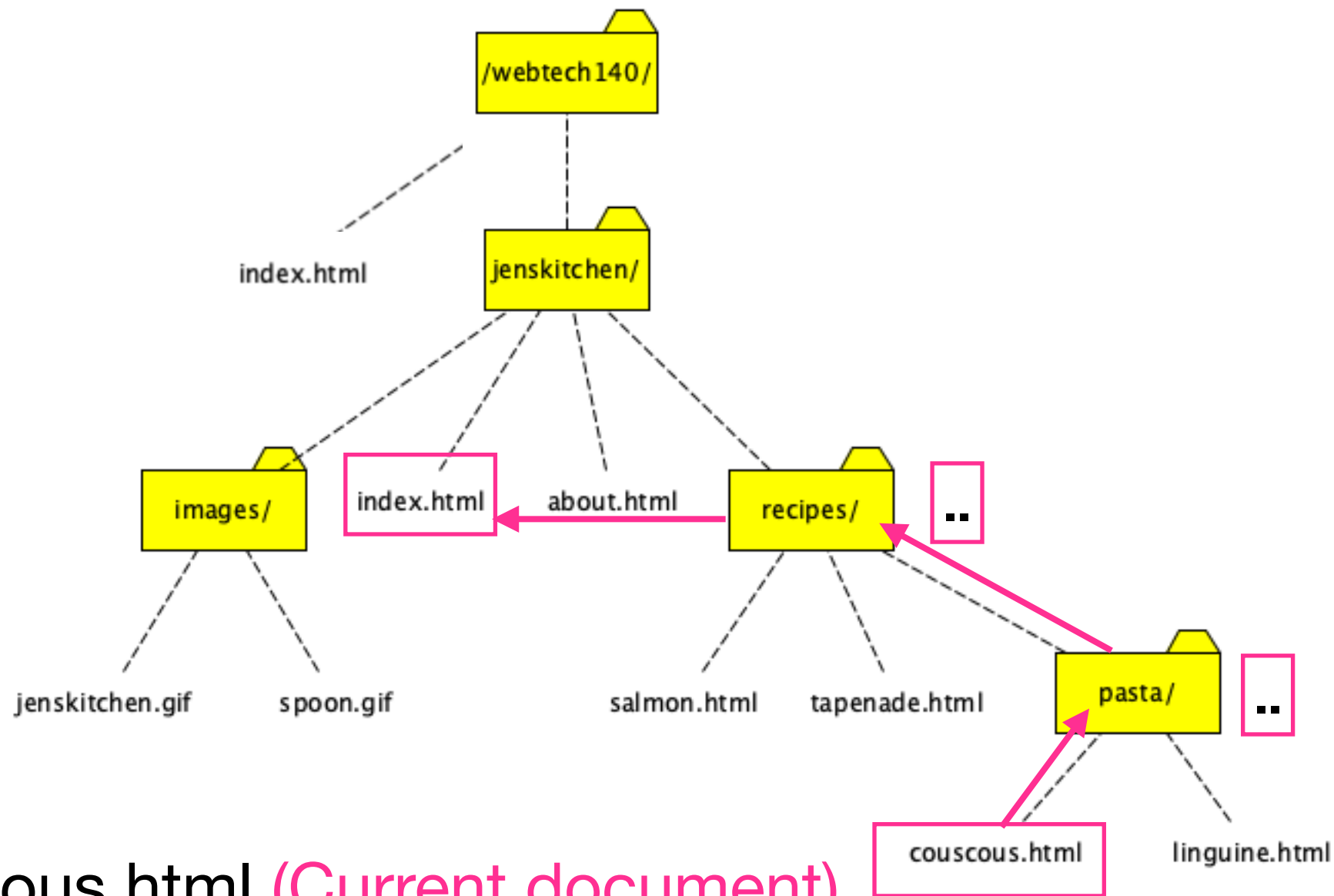
- index.html (Current document)
 - `<a href="recipes/pasta/couscous.html">Couscous with Peas and Mint</a>` OR
 - `<a href="./recipes/pasta/couscous.html">Couscous with Peas and Mint</a>`

Linking to a higher directory (one level)



- salmon.html (Current document)
- `<a href="../index.html">[Back to home page]</a>`

Linking to a higher directory (two levels)

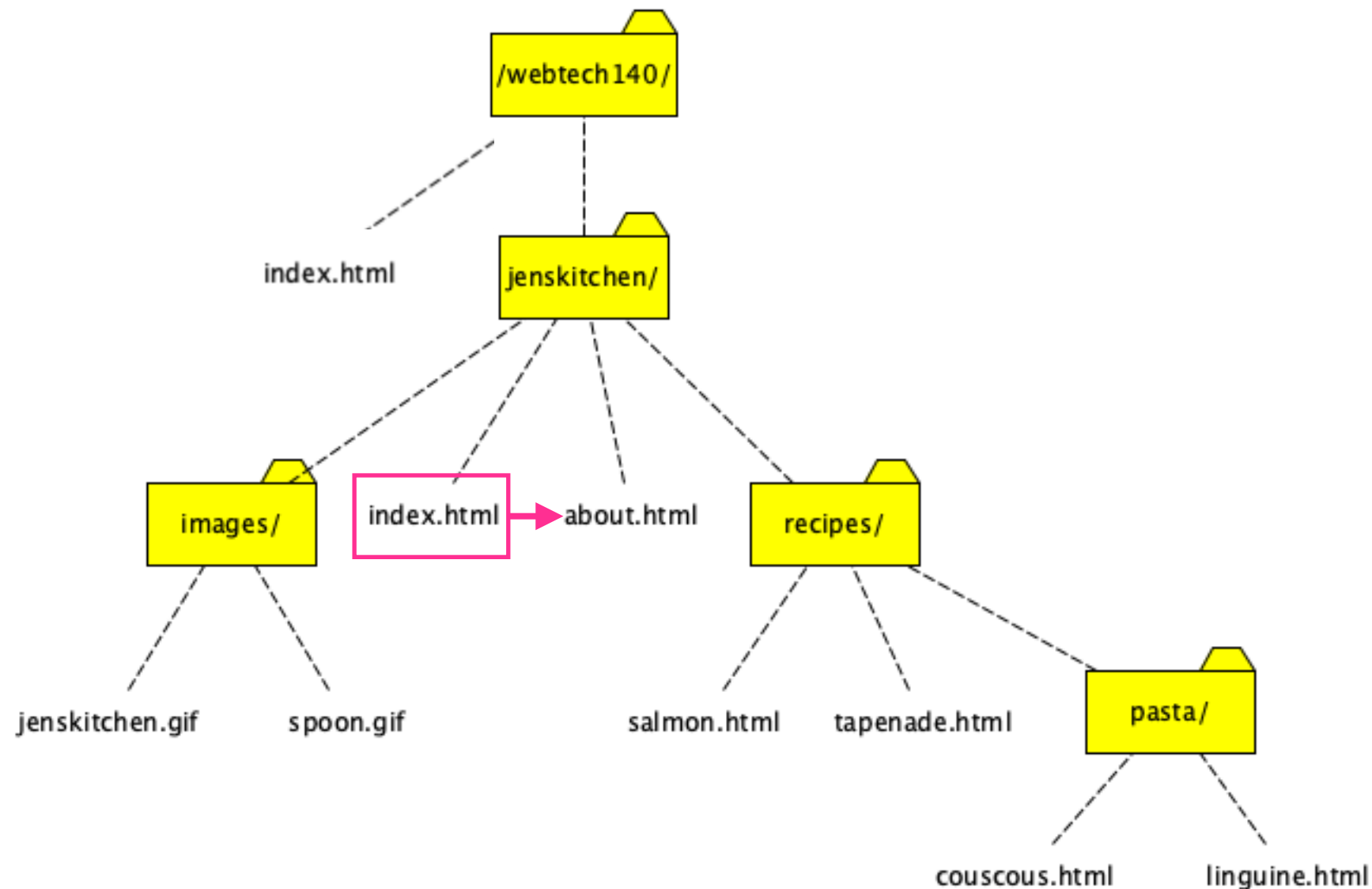


- couscous.html (Current document)
- `<a href="../../index.html">[Back to home page]</a>`

Site root relative pathnames

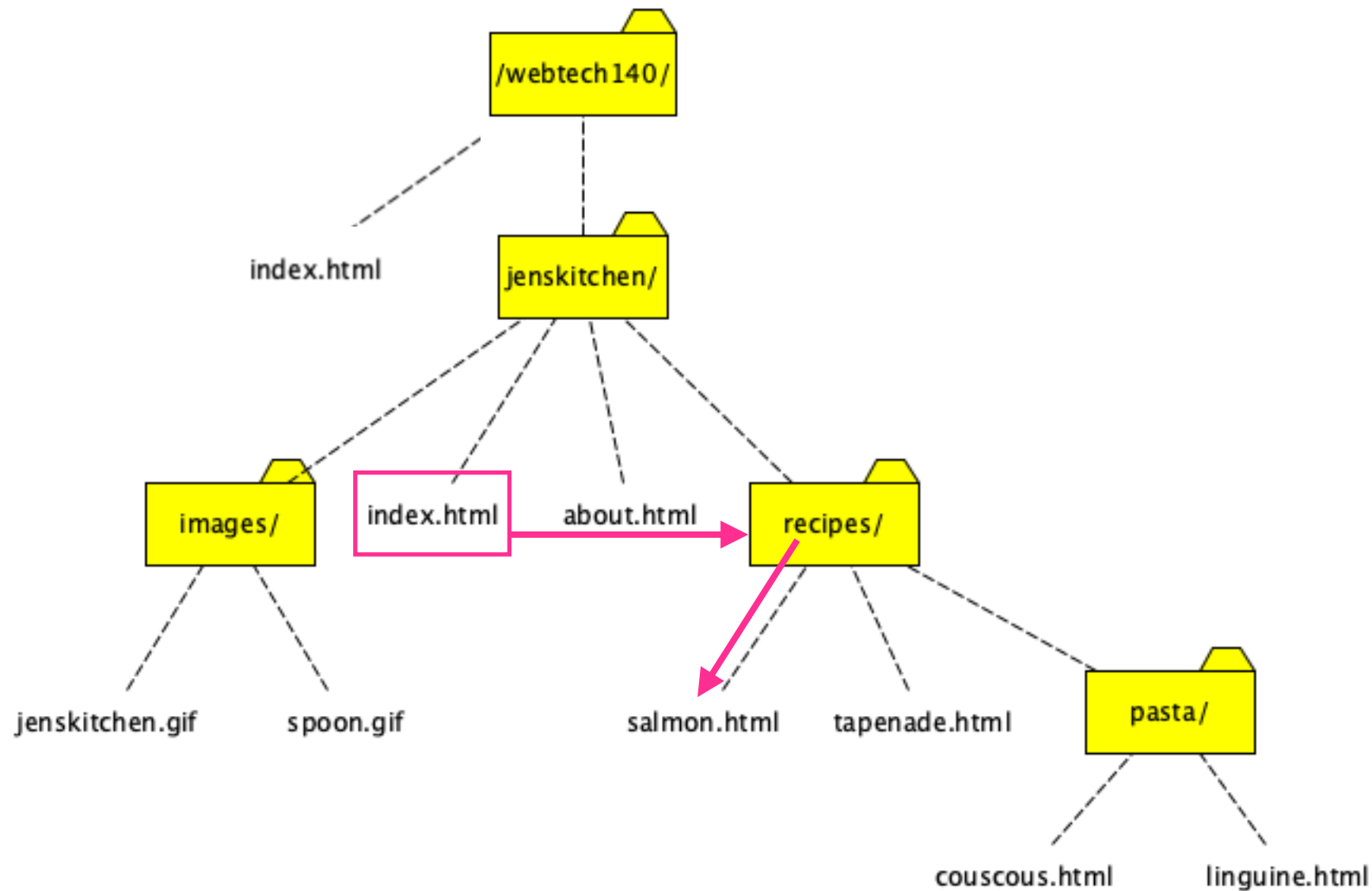
- All websites have a **root directory**
- **A root directory**
 - is the directory that contains all the directories and files for the site.
- All pathnames are **relative to the document** with the link.
- All pathnames can also **start at the root directory** and list the subdirectory names until getting to the file you want to link to (known as site **root relative "/"**).

Linking within a directory



- index.html (Current document)
- `<a href="/jenskitchen/about.html">About this site</a>`

Linking to a lower directory (one level)



- index.html (Current document)
 - `<a href="/jenskitchen/recipes/salmon.html">Garlic Salmon</a>`

Linking to a specific point in a page

- [Common examples](#) of linking to a specific point in a page
 - "[Back to top](#)" links at the bottom of a long page
 - A list of contents on a page that takes the user to the relevant section of that page
 - Links within text to footnotes or definitions
- The [destination anchor](#) allows the page author to [mark](#) specific points in a page that a [source anchor](#) can point to.

Linking to a specific point in a page

- Two-part process

1. Identifying the destination

- use `<a>` element with the `id` attribute and a value to the `id` attribute.

2. Linking to the destination

- use `<a>` element to make a link to the destination with the `href` attribute. The value of `href` attribute should be the same as the value of `id` attribute, preceded by a pound or hash sign (`#`).

Linking to a specific point in a page

Anchor text **id="top"**

The anchor text, link label, link text, or link title is the visible, clickable text in a hyperlink. The words contained in the anchor text can determine the ranking that the page will receive by search engines. Since 1998, some web browsers have added the ability to show a tooltip for a hyperlink before it is selected. Not all links have anchor texts because it may be obvious where the link will lead due to the context in which it is used. Anchor texts normally remain below 50 characters. Different browsers will display anchor texts differently. Usually, web search engines analyze anchor text from hyperlinks on web pages. Other services apply the basic principles of anchor text analysis as well. For instance, academic search engines may use citation context to classify academic articles,[1] and anchor text from documents linked in mind maps may be used too.[2]

Contents

1. [Overview](#)
2. [Search engine algorithms](#)
3. [Terminology](#)

Overview **id="overview"**

Anchor text usually gives the user relevant descriptive or contextual information about the content of the link's destination. The anchor text may or may not be related to the actual text of the URL of the link. For example, a hyperlink to the English-language Wikipedia's homepage might take this form:

The anchor text in this example is "Wikipedia"; the longer, but vital, URL http://en.wikipedia.org/wiki/Main_Page needed to locate the target page, displays on the web page as Wikipedia, contributing to clean, easy-to-read text.

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Search engine algorithms **id="search"**

Anchor text is weighted (ranked) highly in search engine algorithms, because the linked text is usually relevant to the landing page. The objective

Moving forward, Google would be paying more attention to a diversified link profile which has a mix of anchor text and other types of links. .[8]

However a 2016 study of anchor text influence across 16,000 keywords found that presence of exact and partial match anchor links continues to have a strong correlation with Google rankings.

August 2016 study conducted by Moz, found that Exact and partial match domains can be affected by over optimization penalty since Google considers domain Brand and naked URL links as Exact match.[9]

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Terminology **id="terminology"**

There are different classifications of anchor text that are used within the search engine optimization community such as the following:

Exact Match

an anchor that is used with a keyword that mirrors the page that is being linked to. Example: "search engine optimization" is an exact match anchor because it's linking to a page about "search engine optimization".

Partial Match

an anchor that is used with a keyword and a variation that mirrors the page that is being linked to. Example: "different search engine optimization techniques" is a partial match anchor when it's linking to a page about "search engine optimization".

Branded

a brand that is used as the anchor. "Wikipedia" is a branded anchor text.

Naked Link

a URL that is used as an anchor. "www.wikipedia.com" is a naked link anchor.

Generic

a generic word or phrase that is used as the anchor. "Click here" is a generic anchor. Other variations may include "go here", "visit this website", etc.

Images

whenever an image is linked, Google will use the "ALT" tag as the anchor text.

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Demo

anchor.html

```
<h4>Contents</h4>
<ol>
  <li><a href="#overview">Overview</a></li>
  <li><a href="#search">Search engine algorithms</a></li>
  <li><a href="#terminology">Terminology</a></li>
</ol>
...
...
...
<h2><a id ="overview">Overview</a></h2>
...
...
...
<a href="#top">[Back to top]</a>
<h2><a id ="search">Search engine algorithms</a></h2>
...
...
...
<a href="#top">[Back to top]</a>
<h2><a id ="terminology">Terminology</a></h2>
```

Demo

<a> : target attribute

- The **target** attribute specifies where to open the linked document.

_blank Open the linked document in a new window or tab

_self Open the linked document in the same window or tab as it was clicked (default)

_parent Open the linked document in the parent frame

_top Open the linked document in the full body of the window

framename Open the linked document in a named frame

`<a href="https://www.sit.kmutt.ac.th" target="_blank">SIT home page</a>`